

larger than for the large cap order size is that the price of the small cap stock is about one-half the price of large cap stocks and small cap volume is about one-tenth large cap stock. Thus, resulting in small cap order sizes that are dramatically larger than for large cap stocks for the same dollar value. The corresponding cost estimate of \$10 million USD was 14.1bp for the large cap stock and 65.1bp for the small cap stock. This results in a small cap stock cost that is 4.6 times greater than for the equivalent dollar value invested in the large cap stock.

Exchange Traded Funds (ETFs). Cost estimates for the liquid ETF were approximately -80% less than costs for large cap stocks. This reduction in cost was primarily due to the shadow liquidity and factor exposure liquidity corresponding to exchange traded funds. For example, using a single stock market impact model, the estimated cost for transacting 100% ADV of a broad market ETF such as the SPY could be as high as 200bp. But the actual trading cost for this investment is closer to 10–20bp since the investor has many options to achieve this market exposure and the desired ETF. Investors could 1) purchase the desired ETF, 2) they could purchase the underlying stocks and create the ETF, 3) they could purchase a Futures contract, exchange the futures for the physical stock, and then create the ETF. Costs corresponding to the illiquid ETFs were found to be -25% to -35% less than large cap stocks.

Futures. Future trading costs (stock index futures) were found to be -80% less than large cap stocks. Future contracts were also found to be less sensitive to larger order sizes than for stocks or for ETFs. Investors could improve the model forecasting accuracy by fine tuning the impact model for each index individually. Portfolio managers wishing to invest in a futures portfolio, however, will incur an incremental cost at the time of futures expiration where they will have to settle the contract then purchase another futures contract. This is known as the “roll cost” and it plays a large part in the total trading cost of futures contracts.

Bonds. The corresponding cost of government bonds were -90% lower than the cost of large cap stocks. Most of the cost was due to the spread cost of the bonds. Unlike equities, where investors can purchase shares at the bid and sell shares at the ask (offer), investors are much more likely to pay the full spread cost when transacting government bonds. The corresponding cost of corporate bonds was dramatically higher than large cap stocks. Our analysis found that the cost of transacting corporate bonds was +526% higher than the cost of transacting large cap stocks. That is, corporate bonds were 6.26 times more expensive to transact than large cap stocks! This appears to be due to much smaller availability of